Silent Corruption in Clinical AI

At 100 dpi: 88.9% of characters read correctly, 32.3% of laboratory values misread

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code, data & harness

ONE POTASSIUM VALUE, ONE REPORT, RESCANNED AT 100 dpi

5.8 → 58.0 mEq/L

FLAG: HIGH
before and after — unchanged

Tesseract returned **"Potassium 58 mEq/L 3.5-5.0 H"** — the reference range survived intact; the decimal point did not. Observed in the single-document degradation sweep: an illustrative case, not a frequency.

Background

- Clinical records reach AI systems as **images** — scans, faxes, photos.
- OCR can preserve almost every character and still corrupt a number.
- Retrieval shows **where** a number came from, not whether it was read correctly.
- A wrong value can carry the **same abnormality flag** as the true one.
- A physiological plausibility bound caught **55.3%** of misreads here.

Methodology

- Evaluated the **image-to-value path** of a RAG clinical decision support prototype.
- Injected controlled image faults into synthetic reports with exact ground truth for **13 analytes**.
- **Experiment 1** — one report, 32 image conditions, **832** observations.
- **Experiment 2** — **10 reports** × 7 resolutions = **910** observations. The **document** is the experimental unit.
- **Calibration gate**: no result unless the clean render returns 13/13.

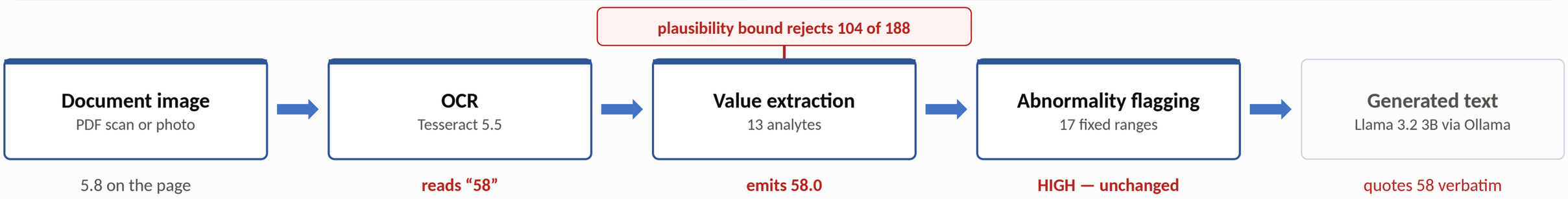


Figure 1. Where the number breaks. Blue stages were exercised with the application's own code and measured here. The potassium example is one observed case from the single-document degradation sweep, not a typical outcome. At 100 dpi the corrupted value was quoted verbatim in 5 of 6 generations; whether it changed a clinical recommendation was not tested.

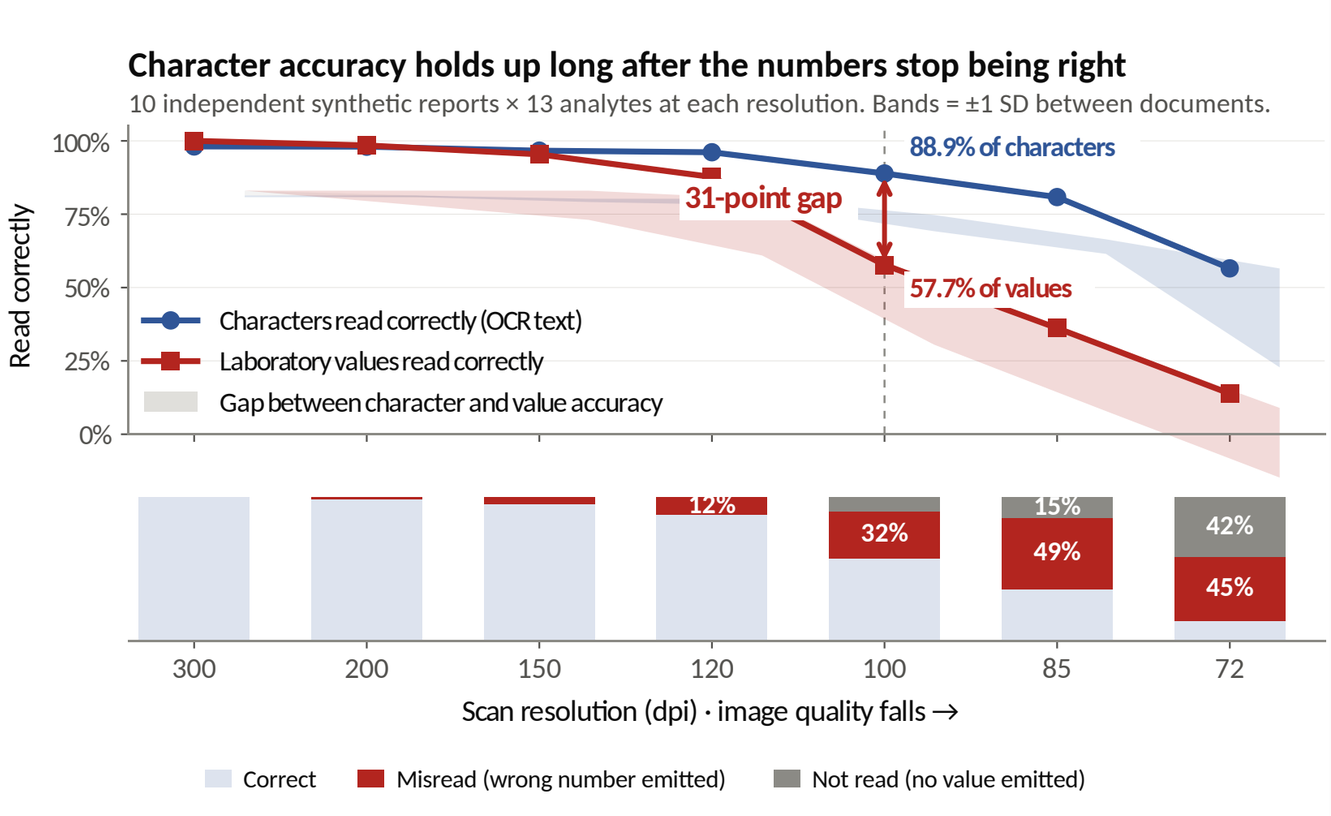


Figure 2. Character accuracy against value accuracy over seven scan resolutions. n = 10 independent synthetic reports × 13 analytes per resolution; the document is the experimental unit. Bands are ±1 SD between documents, not confidence intervals. *Misread* (a wrong number was emitted) and *not read* (no value was emitted) are different failures and are never pooled. Misreading falls between 85 and 72 dpi because values stop being read at all. Resolution, font size and pixels per glyph are entangled: this does not isolate a cause.

Results

1. High character accuracy can coexist with numerical-value errors.

At 300 dpi not one value was misread in any of the 10 reports. At 100 dpi **88.9%** of characters were read correctly while **32.3%** of values were misread — a wrong number emitted — and a further **10.0%** were not read at all.

2. Nearly half of corrupted values kept the true flag.

89 of 188 (47.3%) carried the ground-truth abnormality flag. The plausibility bound rejected 104 of 188 misreads and made **33 of those 89** detectable. It rejected 0 of 636 correct values — but the generator **guarantees** that, so it is not a false-positive rate.

3. The LLM extractor repaired nothing (one document).

LLM extractor: Llama 3.2 3B via Ollama (temperature 0). On identical OCR text both extractors were correct on the **same 188 of 260** values: **0 repairs, 0 new errors**. On 4 of 24 runs the LLM returned no values at all after malformed JSON.

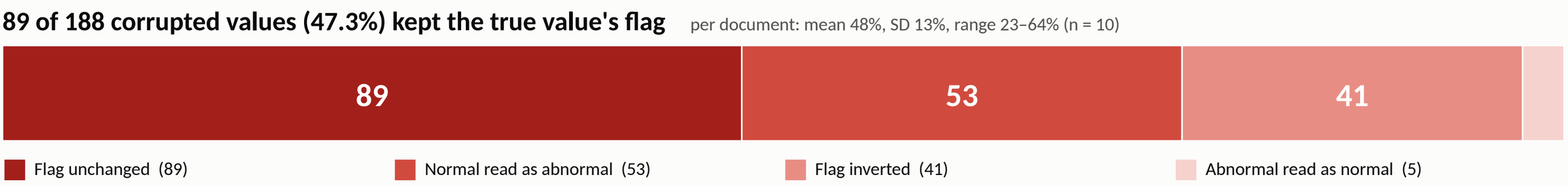


Figure 3. What the flagging layer did with each of the 188 corrupted values, pooled over the 10 reports. *Flag unchanged* means the extracted value carried the same abnormality flag as the true value; it does not establish that no other component could detect the error — the plausibility bound made 33 of these 89 detectable. All 41 inversions reverse direction: 31 low → high, 10 high → low.

Knowing where a number came from is not the same as knowing it was read correctly.

What this does not show. Synthetic reports only — no real patient data, and no clinician has reviewed any of it. 13 analytes, one layout, one OCR engine. Only the resolution sweep is replicated across 10 documents; the degradation sweep and the extractor comparison each use **one**. The LLM extractor is Llama 3.2 3B via Ollama (temperature 0). The safeguard's zero rejections among correct values is **guaranteed by the generator**, which samples inside the plausibility envelope — it is not a real-world false-positive rate. Flags use the system's own fixed ranges, not age- or sex-aware. Retrieval ranking, citation faithfulness and clinical effectiveness were not evaluated. Research prototype — not clinically validated or deployable.

Gao, Y. et al. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv:2312.10997, 2023.

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